POLITECNICO MILANO 1863 | DIPARTIMENTO DI MECCANICA

Project presentation

25.02.2026 | Arturo Giacobbe

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Goal

Teach long manipulation sequences to a robotic arm leveraging **Vision-Language Models (VLMs)**

Inputs:

- Images
- Textual instructions

You are asked to:

1. **Teach** a series of **pick and place** actions to the robot
2. **Validate** the task plan
3. **Test** the robot policy **without human intervention**

You will perform simulations and real-world experiments

Goal

Intended learning objectives

By the end of this project, you will be able to:

1. Understand what a Vision-Language Model is and how it works.
2. Use an API-based model from Hugging Face.
3. Process visual and textual input.
4. Convert high-level instructions (e.g., “pick the red cube”) into robot actions.
5. Execute a task pipeline in simulation.
6. Deploy the task plan to a real ABB GoFa robot.
7. Evaluate limitations and failure cases.

Applications:

- Pick and place
- Sorting objects based on color or shape
- Pick up the requested object

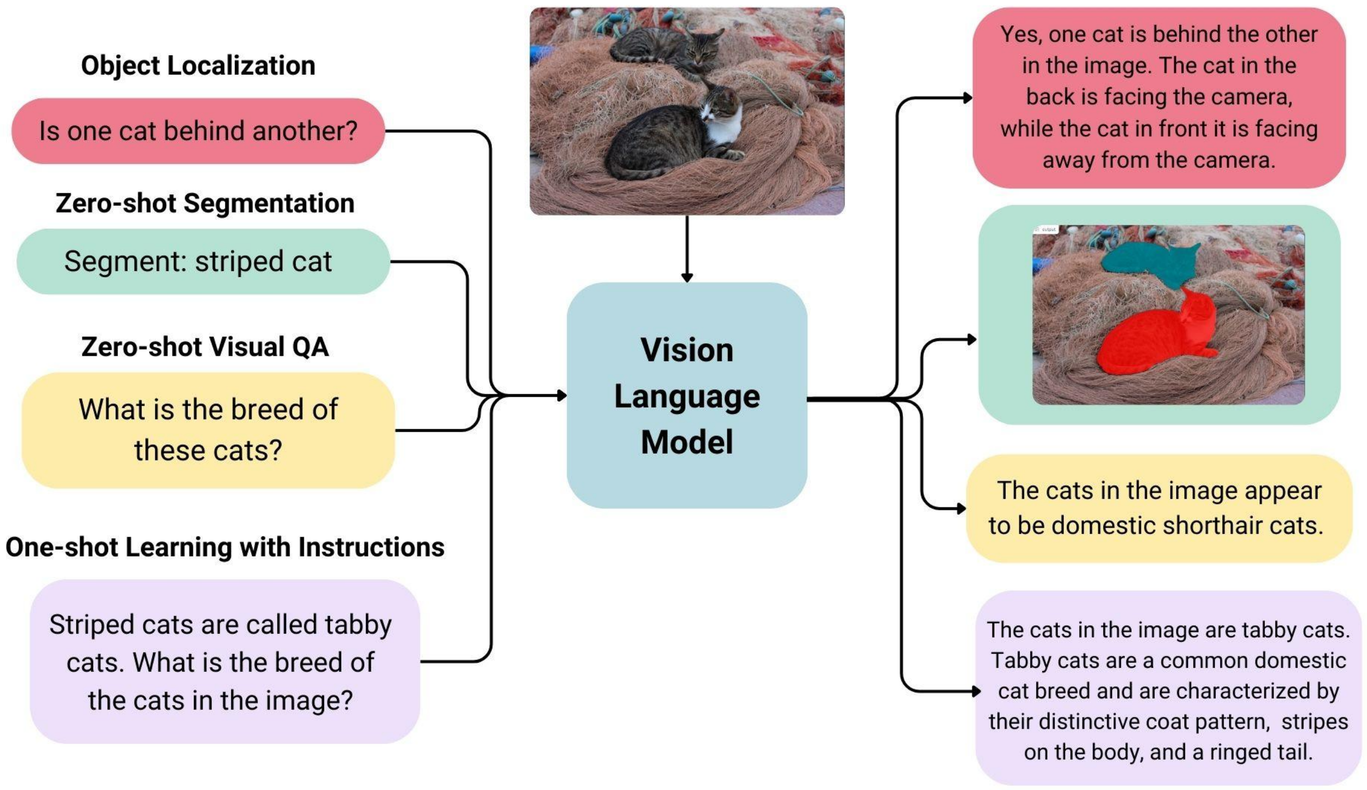
Vision-Language Models

VLMs are a type of **generative models** that take image and text inputs and generate text outputs.

They can be used for many different tasks:

- Image recognition
- Visual question answering
- Document understanding
- Bounding boxes
- Segmentation masks

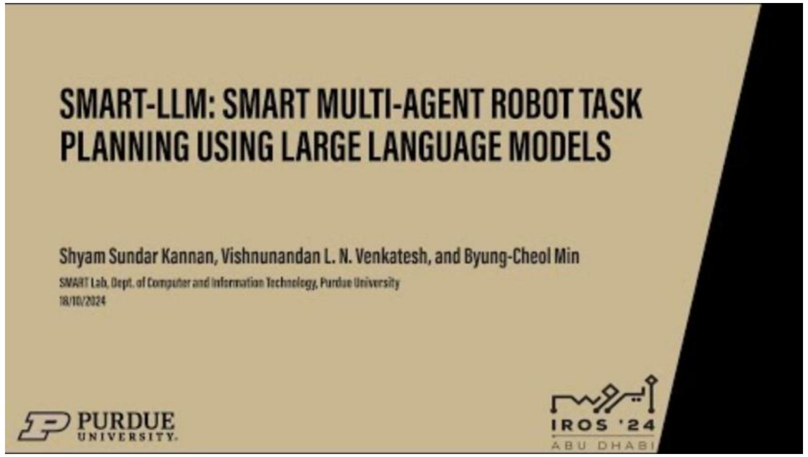
VLMs can also be used to tell a robot **what object** to pick and **where it must be placed**.



https://huggingface.co/blog/vlms

Generative models and task planning

Decompose high-level, abstract goals (e.g., "clean the room") into a **logical**, **sequential** set of steps (e.g., "navigate to X" "pick up X" "deposit X into Y").

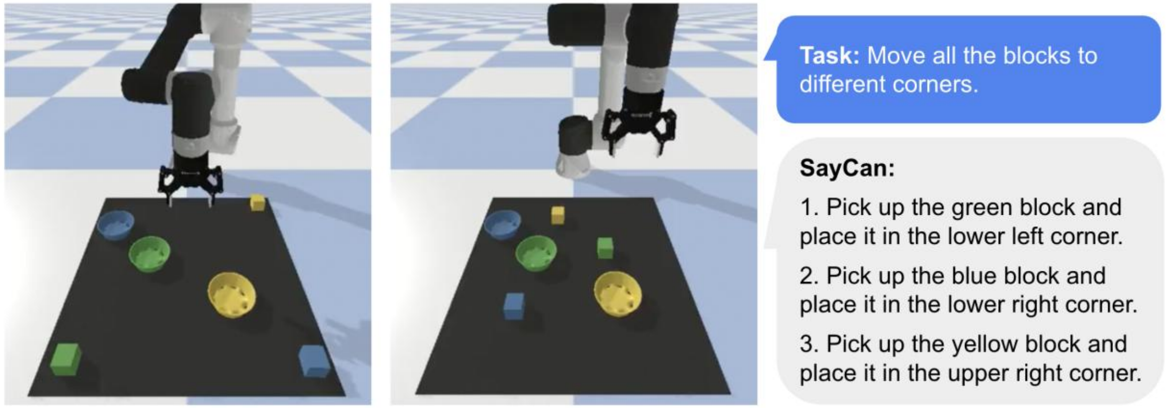


https://www.youtube.com/watch?v=TGNFDeohW6o&t=22s

It tells the robot:

- Where to move (x, y, z)
- In an ordered and logical sequence

Recent applications try to use **VLMs for task planning**



https://say-can.github.io/

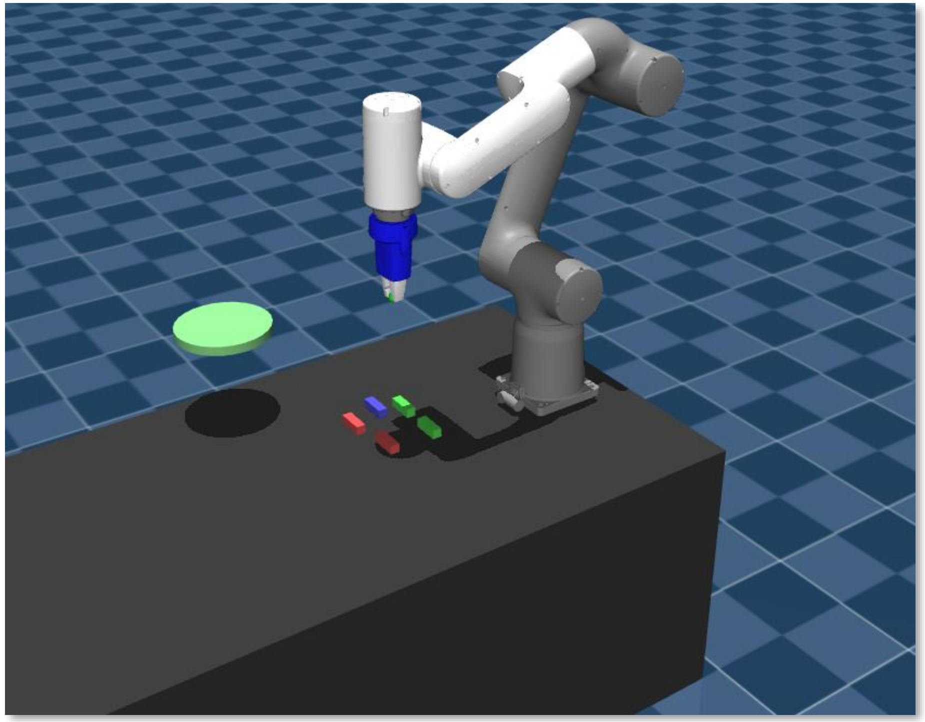
Simulation environment

- MuJoCo environment with the **ABB GoFa robot arm** and various objects on the table.
- It is fundamental to test the robot policy before deploying it to the real world

How to use it?

→ See tutorial on the provided Colab notebook

→ Check MuJoCo documentation



https://mujoco.org/

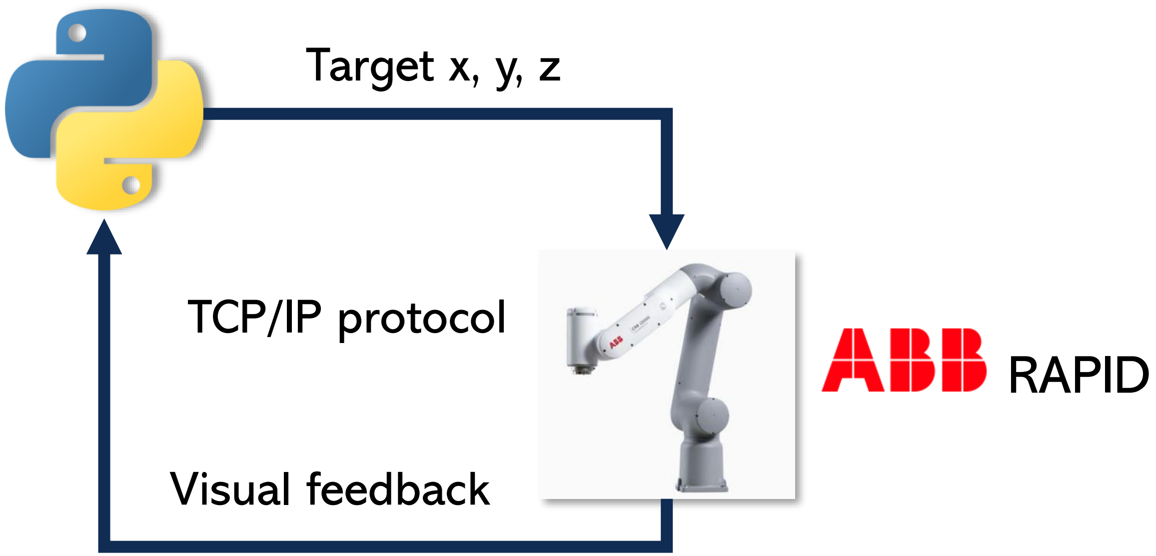
https://github.com/artuurog/GoFa_MuJoCo_sim/tree/main

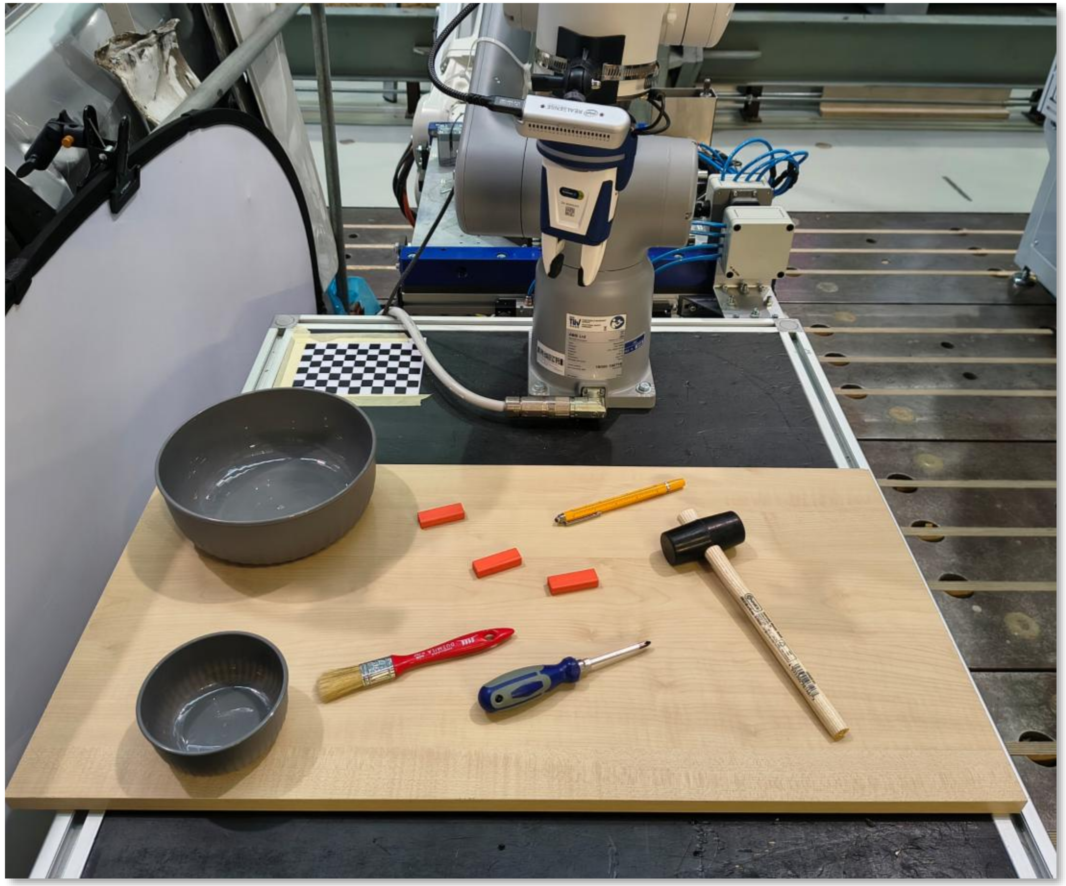
Real robot

Once you are satisfied with the simulations, you can proceed with real world experiments

The robot is an ABB GoFa manipulator arm

Low level control:





Workflow - Training phase

Inputs

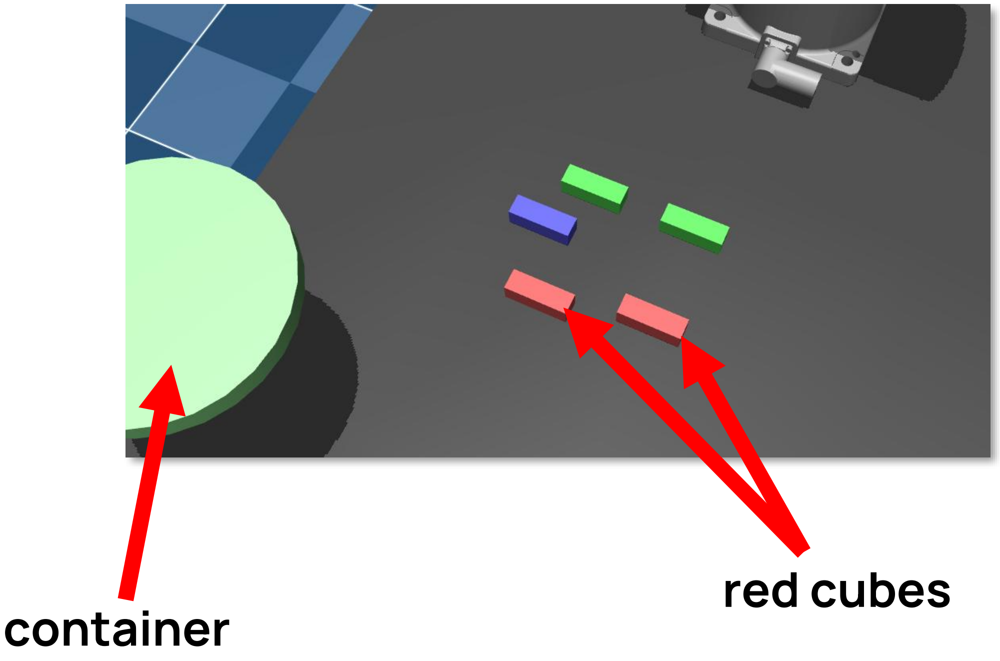
➢ Human instruction (e.g. «Pick the red cube and place it in the container», «clear the table»)
➢ Image of the workspace

Let a Vision-Language Model (VLM) reason about:

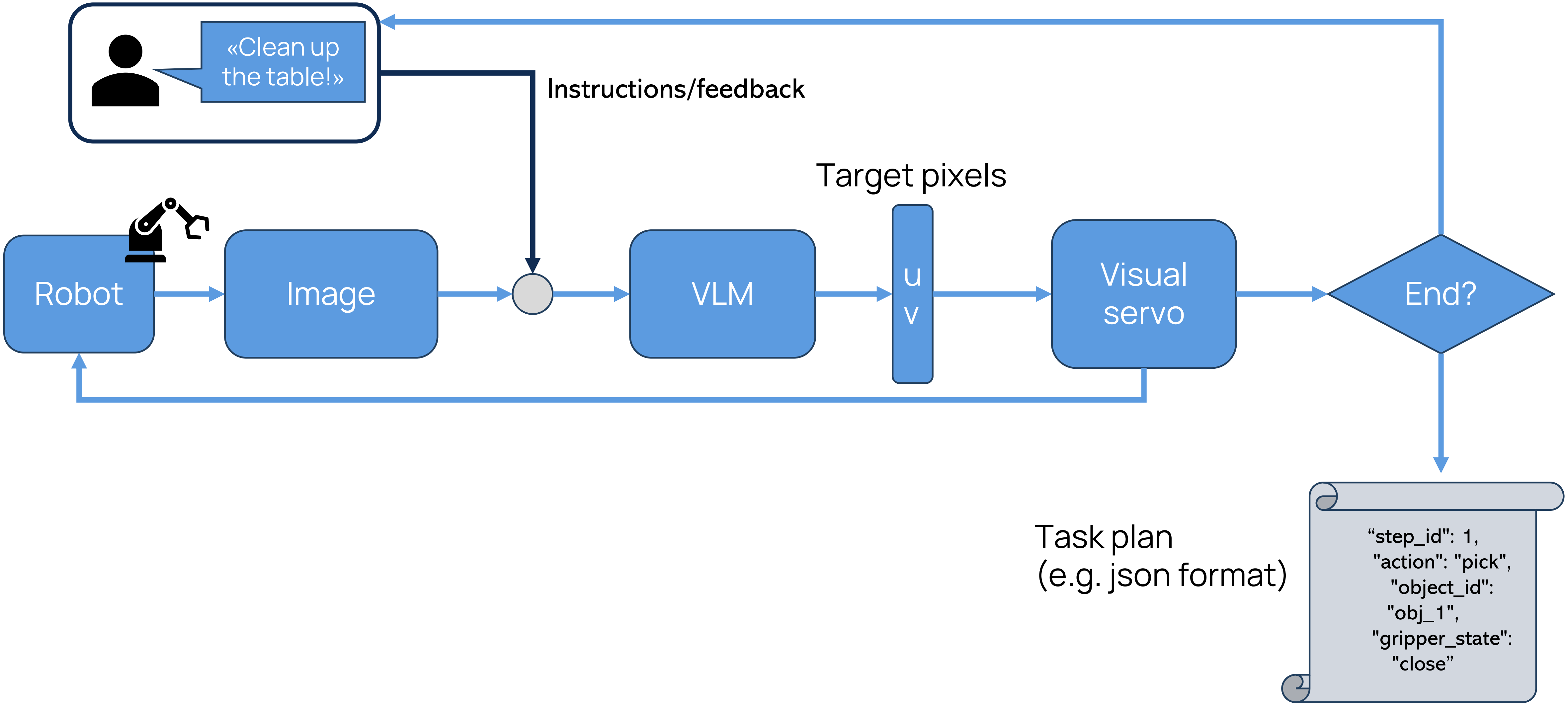
1. Decompose task into subtasks
2. Identify the target to reach
3. Save the operation to a task plan file (e.g. json file)

The human:

➢ Gives textual instructions to the robot
➢ Provides corrective feedback if needed



Training phase



Test phase

Inputs:

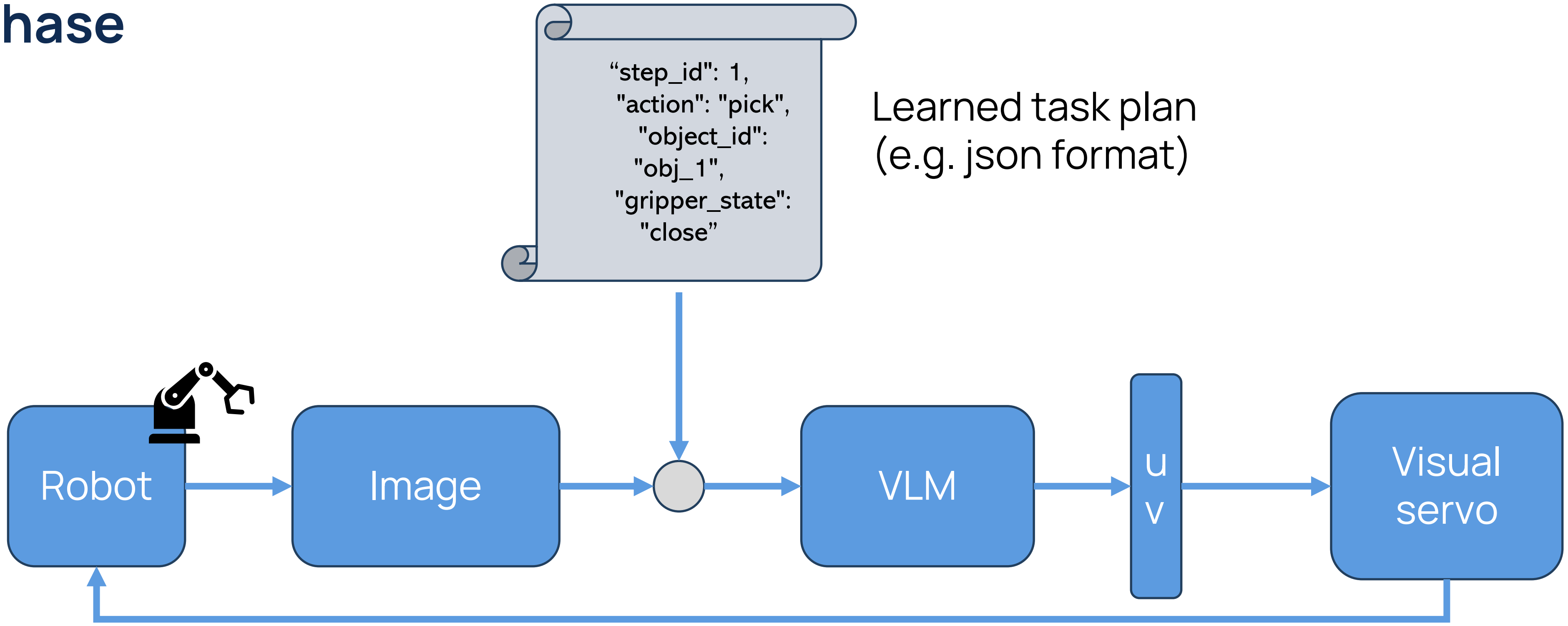
- Task plan
- Image of the workspace

The robot:

- Takes a picture of the objects on the workspace
- Gives the learned task plan + image to the VLM
- Uses a VLM to infer the object to pick and where to place it

Autonomous operation

Test phase



The robot controller has to:

- Understand the step of the operation based on images, robot states…
- Localize the target to reach
- Open or close the gripper?
- Has the robot reached the target?

Robustness analysis

Object detection

- Is the VLM able to correctly locate the objects in the image? Is an additional object detection module needed?

Spatial precision

- Is the VLM precise enough to guide the robot? Do you need scaffolding or visual prompting for better guidance?

Reasoning ability of the models

- Is the VLM able to elaborate and decompose complex instructions? Can it elaborate implicit requests (e.g. «I'm thirsty», «Mount the wheel on the shaft»)?

Sim-to-real

- If I learn one plan in simulation, can I use it as it is in a real-world scenario?

Robustness analysis

Models

- Compare the performance of different models and sizes.
- Which one can better interpret implicit requests?
- Which one generates the most detailed robot instructions?

Robust control loop

- What happens if you move objects suddenly during the test?
- Does the robot update its target?
- Is it able to adapt in real time to unexpected situations?

References and Materials

Visual servo: **https://faculty.cc.gatech.edu/~seth/ResPages/pdfs/HutHagCor96.pdf**

MuJoCo documentation: **https://mujoco.readthedocs.io/en/stable/overview.html**

HuggingFace: **https://huggingface.co/**

Github repo: **https://github.com/artuurog/Machine_Learning_LAB**

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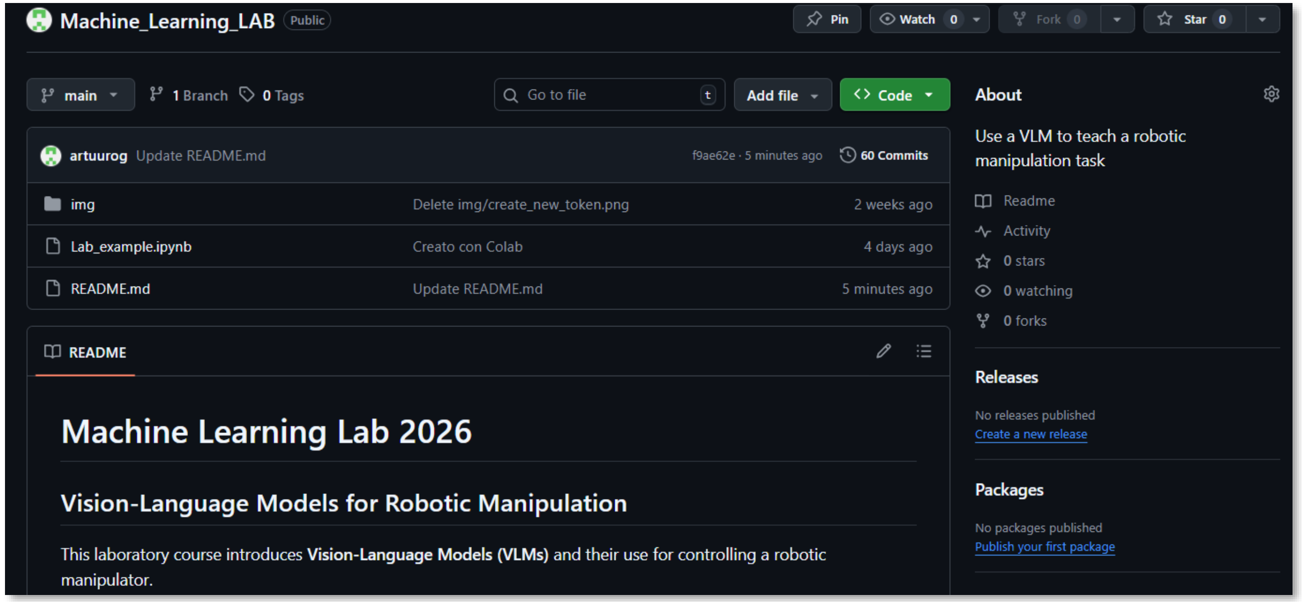
Github repo

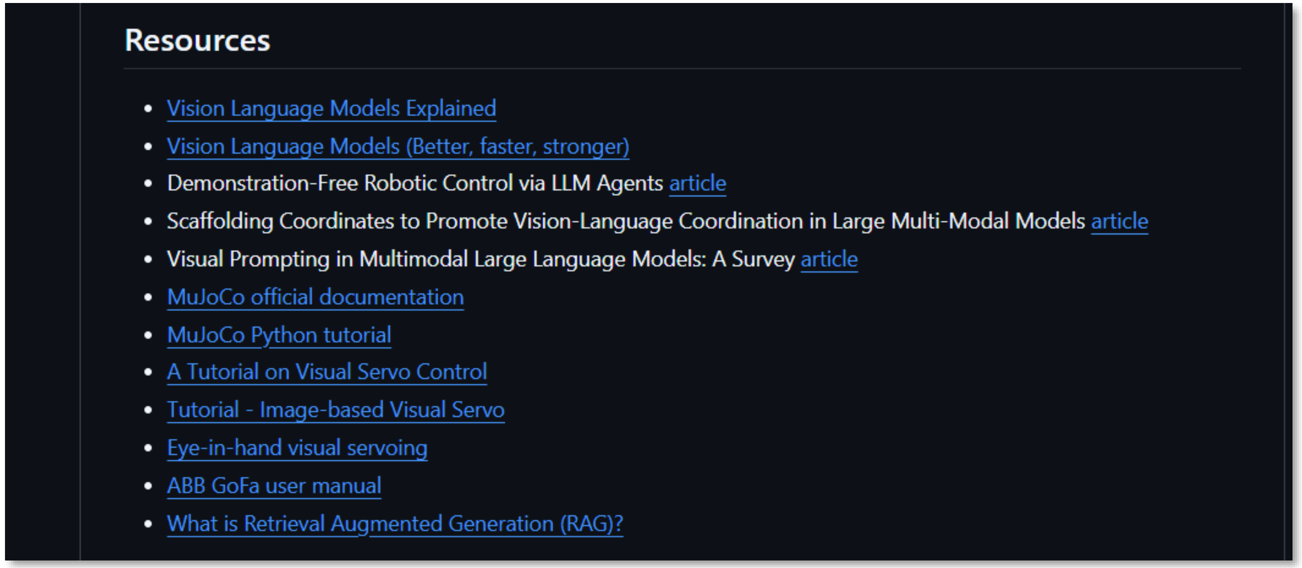
https://github.com/artuurog/Machine_Learning_LAB

Here you can find a more detailed description of the project

The repository contains:

- Colab notebook (.ipynb file) with a brief tutorial
- Guides and manuals
- External resources
- Useful links
- Articles to start your research





Thank you!